

SOBRE B - TECHNICAL PROPOSAL

Company: QuantumNet Solutions SL

Tender: CTTI-2026-36

Subministrament, instal·lació i posada en producció d'equipaments de distribució de clau quàntica

EXECUTIVE SUMMARY

QuantumNet Solutions SL presents this technical offer in response to CTTI tender CTTI-2026-36 for the supply, installation and commissioning of quantum key distribution equipment. Our proposal is grounded in direct, hands-on experience delivering operational QKD infrastructure across two completed public-sector pilots in Catalonia and the Basque Country, and we are confident it represents the most technically capable and operationally mature response to this procurement.

PROJECT TEAM AND QUALIFICATIONS

The project will be led by Dr. Jordi Puig, who will serve as Programme Manager for the full contract duration. Dr. Puig holds seven years of optical network experience and has spent the last three years focused exclusively on CV-QKD deployment engagements. He holds CCDP certification and served as lead project manager on both of the public-sector QKD pilots referenced above, overseeing node deployment, acceptance testing and handover at each site. His direct familiarity with the challenges of integrating QKD equipment into live operational DWDM environments is a material differentiator for this contract.

Solution architecture will be the responsibility of Maria Solà, a network architect with six years designing and delivering complex optical infrastructure. For the past two years, Maria has been embedded within the Telefonica Barcelona research node, leading the integration of QKD equipment into their carrier-grade DWDM backbone. She holds CCDP certification and brings direct experience aligning QKD performance with production traffic management constraints - precisely the scenario CTTI faces across its node topology.

The deployment team comprises two network administrators, both holding CCNP certification with three years of experience in QKD-adjacent optical transport roles. Day-to-day operations and Level 3 escalation support will be provided by a CCIE-certified senior engineer who holds formal manufacturer accreditation from our QKD equipment vendor. This engineer is committed to a minimum of 120 hours of dedicated support engagement across the deployment and stabilisation phases, with response SLAs defined contractually.

Security engineering for KMS integration and cryptographic configuration will be handled by a dedicated security engineer with four years of applied cryptography experience, specialising in KMS/HSM deployment. This profile holds ISO 27001 Lead Implementer certification and has directly configured FIPS 140-3 validated HSM environments for financial and government clients.

The proposed CV-QKD system delivers a minimum net symmetric key generation rate of 1.5 kbps per emitter/receiver pair at 10 dB optical loss - exceeding the minimum 1 kbps requirement - with a security parameter of $\epsilon = 10^{-10}$, an order of magnitude tighter than the required 10^{-9} .

The KMS component is FIPS 140-3 certified and fully compliant with ETSI GS QKD 004 and ETSI GS QKD 014, including all key delivery methods. All equipment ships in standard 19-inch rack format at 2U per main unit with a maximum depth of 550mm.

A key differentiator of our equipment offer is full DWDM-to-dark-fibre reconfiguration capability via software-defined wavelength adjustment, requiring no hardware replacement when network topology changes. This directly addresses the flexibility requirement cited in Criterion 1.2 regarding adaptation to changes between DWDM and dark fibre connectivity.

Resilience design includes dual redundant power supplies on all primary units and automatic failover with a recovery time of 30 seconds. Critical optical components are dual-sourced from two independent manufacturers, eliminating single-vendor dependency at the component level. Each emitter and receiver can be replaced independently; factory-paired units that require full system replacement are explicitly absent from our offer.

DEPLOYMENT RISK MANAGEMENT

Our risk analysis for the deployment phase has identified the following scenario as the primary technical risk warranting a specific mitigation plan:

Risk 1 - Optical signal degradation at legacy fibre nodes: Several nodes in the target topology may present aged or non-characterised fibre runs. Degraded physical layer performance at these nodes could compress the quantum channel margin below the operational threshold. Our mitigation is a structured pre-installation OTDR survey at each node, completed a minimum of three weeks before equipment delivery, with a defined acceptance criterion for fibre qualification. Nodes failing qualification will trigger a corrective fibre assessment before any QKD equipment is commissioned at that site.

MONITORING AND SERVICE VISIBILITY

A fully operational NOC dashboard will be live from day one of service, not at a later milestone. The dashboard presents Secure Key Rate, excess noise, optical loss, and received signal power per node in real time, refreshed on a five-minute cycle. All primary KPIs defined in Section 3.1.5 of the PPT are instrumented, including SKR thresholds, tampering alerts and hardware health indicators. The platform integrates natively with CTTI SNMP and HTTP management infrastructure. Alerting is automated with configurable threshold-based notifications routed to both the QuantumNet NOC and CTTI monitoring teams simultaneously.

A dedicated two-person data analysis team with specific quantum network traffic analysis experience will be assigned to the data analysis phase. Both analysts have worked previously on traffic characterisation in QKD-enabled optical networks and are familiar with the interplay between quantum channel performance metrics and classical traffic coexistence. Deliverables will be structured according to the templates defined in PPT Section 5.4.

DOCUMENTATION AND HANDOVER

Technical manuals will be delivered for all supplied equipment. Documentation packages will be provided in Catalan as required by Section 3.1.2 of the PPT. Full handover documentation will be available at the close of the service.

